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Efficacy and safety of high-dose hypofractionated palliative radiotherapy (40 Gy in 10 daily fractions) for non-small cell lung cancer

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Purpose: Palliative thoracic radiotherapy plays a crucial role in the management of patients with non-small cell lung cancer (NSCLC). Conventional dose-fractionation schemes often yield modest and short-lived tumor responses. This study aims to evaluate the efficacy and safety of a high-dose hypofractionated radiotherapy regimen of 40 Gy in 10 daily fractions in NSCLC.

Materials and Methods: Consecutive NSCLC patients who underwent palliative thoracic radiotherapy using the 40 Gy in 10 daily fractions regimen between January 2017 and December 2022 were identified from institutional databases of two tertiary oncology centers. Treatment responses were assessed using the Response Evaluation Criteria in Solid Tumor criteria. Local tumor control rate was estimated using cumulative incidence function, considering death as a competing event. The rates of radiation pneumonitis and esophagitis were assessed.

Results: Sixty-two eligible patients were included. Median age was 72.6 years, with 52 patients (83.4%) having Eastern Cooperative Oncology Group performance status of ≥2. Radiotherapy was given to 33 patients (53.2%) with locally advanced primary NSCLC and 29 patients (47.8%) with oligopersistent or oligoprogressive NSCLC. Fifty-three tumors (85.4%) were considered unsuitable for stereotactic radiotherapy by location or size. With a median follow-up of 4.1 years, the overall response rate was 69.3%. Local control rates at 1 and 3 years were 96.7% and 76.9%, respectively. Grade 1–2 radiation pneumonitis and esophagitis occurred in 46.7% and 8.1% of patients. One case of grade 3 esophagitis was observed.

Conclusion: The 40 Gy in 10 daily fractions regimen demonstrates high local tumor control rate and a favorable safety profile, representing an effective and safe palliative option for NSCLC patients.

Keywords: Non-small cell lung carcinoma, Radiotherapy, Radiotherapy dosage, Palliative care

Introduction

Over half of patients with non-small cell lung cancer (NSCLC) present with locally advanced or metastatic disease [1]. Current practices in conventional palliative radiotherapy for NSCLC vary significantly across clinicians and institutions, with common regimens ranging from 8–10 Gy in a single fraction to 30 Gy delivered in 10 fractions [2,3]. While these regimens are effective in alleviating symptoms, the resulting tumor response is often modest and short-lived. Higher-dose regimens, such as 39 Gy in 13 fractions, has been shown to improve overall survival compared with conventional schedules, at the expense of a slightly higher incidence of radiation-induced esophagitis [4]. Aggregate data from a systematic review also suggest a modest survival advantage survival with

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radiotherapy schedules with biologically effective dose of ≥35 Gy [5]. Evidence for further dose escalation to 50–70 Gy in 10–25 fractions exists, but the studies were primarily focused on patients with early-stage disease with relatively small tumor sizes [6,7].

As the prognosis of NSCLC continues to improve with breakthroughs in molecularly targeted agents and immune checkpoint inhibitors, there is a growing demand for more effective local therapies, driven by the need to optimize local primary tumor control and to address oligometastatic diseases [8]. The advent of stereotactic body radiotherapy (SBRT) has allowed effective ablation of primary tumors and lung metastases. Nevertheless, factors such as tumor size and location may limit its applicability. The general upper limit of tumor diameter considered acceptable for SBRT is 5 cm, delivering high ablative radiation to targets beyond this size limit is dosimetrically challenging [9]. Tumors with endobronchial involvement and patients on anticoagulation are also considered not suitable for SBRT. Given these considerations, there remains a room to explore effective and safe non-SBRT palliative radiotherapy regimens for patients with NSCLC, with an aim to palliate symptoms and attain local tumor control.

Over the past decade, our group employed a high-dose hypofractionated regimen of 40 Gy in 10 daily fractions as palliative radiotherapy for selected patients with NSCLC. This represents a dose escalation from the conventional regimen of 30 Gy in 10 daily fractions. By utilizing hypofractionation at 4 Gy per fraction, we aim to increase the biologically effective dose, potentially enhancing local tumor control while maintaining an acceptable profile of radiation-related toxicities. The brief two-week treatment course is chosen to optimize patient compliance and convenience, including elderly patients and those with suboptimal performance status. The current study reports on the efficacy and toxicity outcomes of this radiotherapy regimen from a jointed cohort of consecutively treated patients.

Materials and Methods

This is a joint retrospective cohort study conducted in two tertiary oncology centers (Queen Elizabeth Hospital and United Christian Hospital, Hong Kong). Approval has been granted by the research ethics committee (reference number: CIRB-2025-101-2), patient consent was waived. Consecutive NSCLC patients who underwent palliative thoracic radiotherapy using the 40 Gy in 10 daily fractions regimen between January 2017 and December 2022 were identified from institutional databases. Patients without cross-sectional images for response assessment were excluded. The disease was staged using American Joint Committee on Cancer/Union for International Cancer Control 8th edition TNM classification.

All patients were treated in the supine position, immobilized using a wing board with bilateral hand grips. Simulation was performed using computed tomography (CT) with intravenous contrast, images were co-registered with positron emission tomography images wherever available. The use of 4-dimensional CT (4DCT) was not mandatory. Gross tumor volume was delineated using lung window. Isocentric expansion of 5–8 mm was used to generate the clinical target volume, followed by anatomic editing according to natural barriers. A 5 mm isotropic planning target volume (PTV) margin was used if 4DCT simulation was in place, otherwise a differential expansion (5–7 mm axially and 10–12 mm superoinferiorly) was used to account for both respiratory motion and set-up error. Radiotherapy plans were optimized either by 3-dimensional conformal radiotherapy (3DCRT) or volumetric modulated arc radiotherapy (VMAT) techniques. The PTV received 40 Gy in 10 daily fractions, with treatment delivered five fractions per week. The organs at-risk included lungs, heart, esophagus, spinal cord, and brachial plexus. Their respective dose constraints were detailed in Supplementary Table S1. Treatment verification was performed using kV orthogonal on-board imaging or cone-beam CT. No concurrent systemic therapy was given during the course of radiotherapy. An example case was illustrated in Fig. 1.

Treatment responses were evaluated by diagnostic CT using the Response Criteria in Solid Tumors (RECIST) 1.1 criteria. Local tumor control rate was estimated using the cumulative incidence function with death assigned as a competing event. A sensitivity analysis was conducted, censoring data at the time of changes in systemic therapy. Subgroup comparisons were conducted using Gray's test. Overall survival (OS) was estimated using the Kaplan-Meier method, calculated from the start of radiotherapy to death or last follow-up. Multivariable analyses were conducted using the Fine-Gray subdistribution hazard model and Cox regression. The rates of radiation pneumonitis and esophagitis were graded in accordance with Common Terminology Criteria for Adverse Events version 6.0. Statistical analyses were conducted using IBM SPSS Statistics version 24.0 (IBM Corp., Armonk, NY, USA) and Stata version 10 (Stata Corp., College Station, TX, USA).

Results

A total of 62 eligible NSCLC patients were identified during the study period, all of whom completed the 40 Gy in 10 daily fractions palliative regimen to the thorax (Table 1). The median age of the cohort was 72.6 years, with 52 patients (83.4%) having Eastern Cooperative Oncology Group performance status of ≥2. Adenocarcinoma (75.8%) and squamous cell carcinoma (17.8%) were the predominant histology types. Actionable alterations were identified

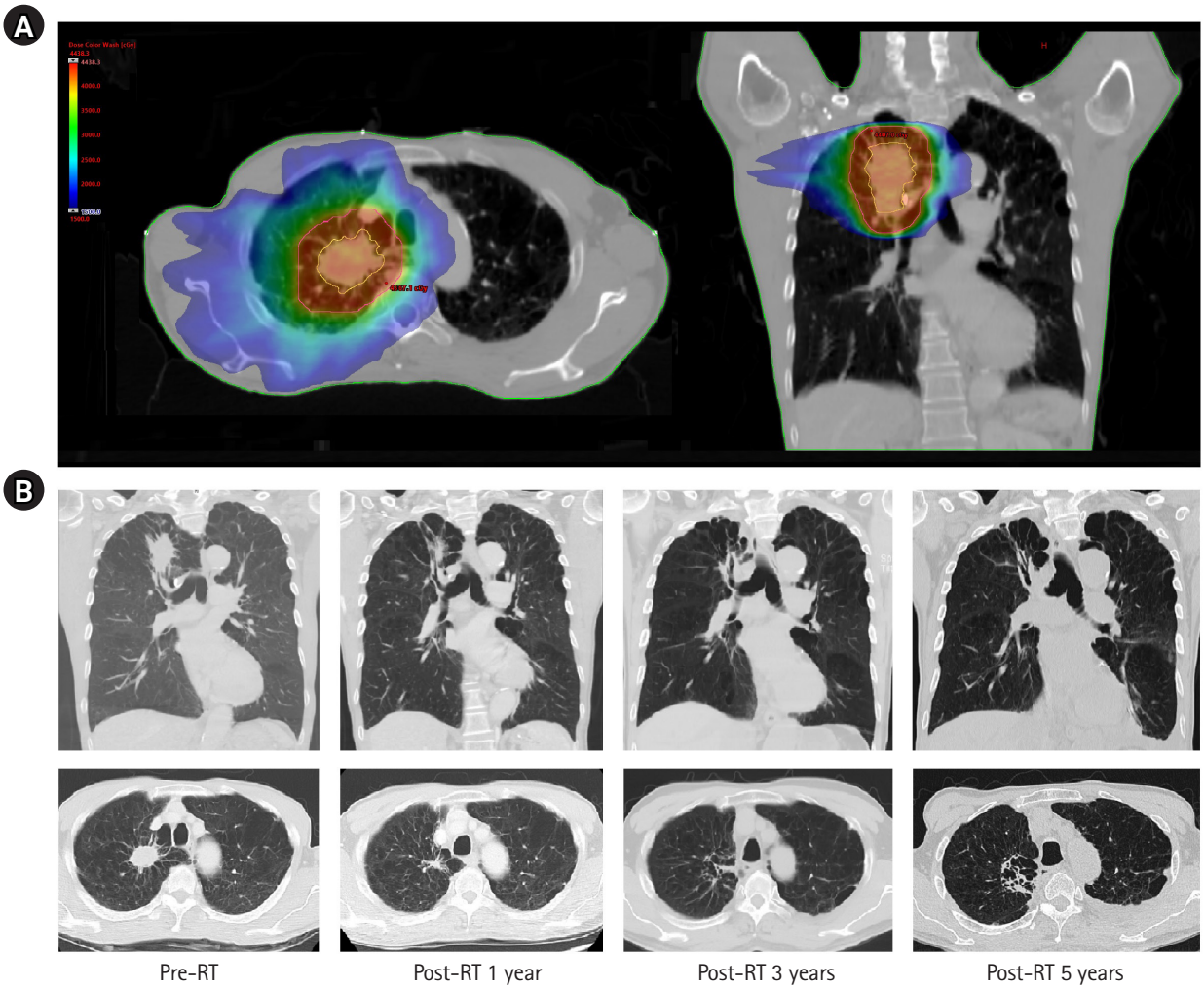

Fig. 1. An example of a radiotherapy (RT) plan using 40 Gy in 10 daily fractions for a 75-year-old man diagnosed with cT2bN0M0 adenocarcinoma of the lung. The 4.7 cm tumor in the right upper lobe was located adjacent to the trachea. (A) Palliative RT was delivered using the volumetric modulated arc radiotherapy technique. The gross tumor volume is indicated in yellow, while the planning target volume is shown in red. The isodose color-wash ranges from 15 Gy to 44.4 Gy. (B) Serial computed tomography images illustrate the continual tumor response and local control over time.

in 50.0% of patients, mostly sensitizing epidermal growth factor receptor mutations. Of the entire cohort, radiotherapy was given to 33 patients (53.2%) with treatment-naïve locally advanced primary NSCLC and 29 patients (47.8%) with oligopersistent or oligoprogressive NSCLC following systemic therapy. Most of the treated tumors were considered unsuitable for SBRT due to large tumor size (n = 34, 54.8%) or ultracentral location (n = 19, 30.6%) with respect to the trachea or the proximal bronchial tree. In the cohort, 29.0% (18/62) of the patients had a change in systemic therapy after radiotherapy, at a median interval of 4.8 months. The distribution of subsequent systemic therapies were tyrosine kinase inhibitors (n = 8), chemotherapy (n = 6), immunotherapy (n = 3), and chemo-immunotherapy (n = 1).

Radiotherapy was delivered using VMAT and 3DCRT in 54 patients (87.1%) and eight patients (12.9%), respectively. 4DCT simulation was employed in 39 patients (62.9%). The median PTV was 168.8 cm^3 (interquartile range [IQR], 100.1 to 266.5). The median PTV V100% was 98.8% (IQR, 97.4% to 99.6%) with a median Dmax of 44.0 Gy (IQR, 43.2 to 44.6). Radiation dose-volume data of the organs at-risk (OARs) were summarized in Table 2. The median lung and heart doses were 6.6 Gy (IQR, 5.4 to 8.8) and 2.2 Gy (IQR, 0.7 to 6.9), respectively. Pre-specified radiation dose constraints were fulfilled in all radiotherapy plans.

With a median follow-up of 4.1 years, the overall response rate of the 40 Gy in 10 daily fractions palliative regimen was 69.3%. The response rates for locally advanced primary tumor and oligometastatic diseases were 78.8% (26/33) and 58.6% (17/29), respectively (p = 0.086). The local control rates at 1-year and 3-year

Table 1. Demographics and treatment details

Characteristic	Value
Age (year)	72.6 (66.8–80.3)
Sex	
Male	36 (58.1)
Female	26 (41.9)
ECOG performance status	
0–1	10 (16.1)
2	45 (72.6)
3	7 (11.3)
Smoking status	
Never smoker	29 (46.8)
Current/ex-smoker	33 (53.2)
Histology	
Adenocarcinoma	47 (75.8)
Squamous cell carcinoma	11 (17.8)
Lymphoepithelioma-like carcinoma	3 (4.8)
Non-small cell lung cancer, NOS	1 (1.6)
Actionable mutations	
EGFR mutation	28 (45.2)
ALK fusion	2 (3.2)
KRAS G12C mutation	1 (1.6)
None	24 (38.7)
Not assessed	7 (11.3)
Indication for radiotherapy	
Local control for primary tumor	33 (53.2)
Oligoprogression	23 (37.1)
Oligopersistence	6 (9.7)
PET-CT before radiotherapy	
Yes	57 (91.9)
No	5 (8.1)
Tumor size (cm)	5.2 (3.4–6.5)
T category[a)]	
T1	8 (12.9)
T2	33 (53.2)
T3	16 (25.8)
T4	5 (8.1)
N category[a)]	
N0	34 (54.8)
N1	8 (12.9)
N2	15 (24.2)
N3	5 (8.1)
M category[a)]	
M0	31 (50.0)
M1	31 (50.0)
Radiotherapy planning technique	
Volumetric modulated arc radiotherapy	54 (87.1)
3-Dimensional conformal radiotherapy	8 (12.9)

Values are presented as median (IQR) or number (%).
ECOG, Eastern Cooperative Oncology Group; NOS, not otherwise specified; EGFR, epidermal growth factor receptor; ALK, anaplastic lymphoma kinase; KRAS, Kirsten rat sarcoma virus; PET-CT, positron emission tomography computed tomography; IQR, interquartile range.
[a)]Cancer stage assessed at the time of palliative radiotherapy.

Table 2. Radiation dose-volume metrics of target volumes and organs at-risk

Structure	Metric	Median (IQR)
GTV	Volume (cm^3)	30.7 (10.0–56.3)
	V100% (%)	100.0 (100.0–100.0)
PTV	Volume (cc)	168.8 (100.1–266.5)
	V100% (%)	98.8 (97.4–99.6)
	Dmax (Gy)	44.0 (43.2–44.6)
Spinal cord	Dmax (Gy)	18.2 (12.6–24.8)
Spinal cord PRV	Dmax (Gy)	22.7 (15.4–30.5)
Total lung (minus GTV)	Dmean (Gy)	6.6 (5.4–8.8)
	V16 Gy (%)	14.3 (10.0–18.9)
	V4.5 Gy (%)	33.8 (22.6–43.2)
Heart	V19.5 Gy (%)	0.6 (0.0–6.3)
	Dmean (Gy)	2.2 (0.7–6.9)
	V39 Gy (%)	0.0 (0.0–0.2)
Esophagus	Dmean (Gy)	6.0 (3.8–8.5)

IQR, interquartile range; GTV, gross tumor volume; PTV, planning target volume; V100%, the relative volume of structure received 100% of prescription dose; Dmax, maximal dose; PRV, planning organ at risk volume; Dmean, mean dose; Vx Gy, the relative volume of structure received x Gy.

of the whole cohort were 96.7% and 76.9% (sensitivity analysis censoring data at the time of changes in systemic therapy, 96.7% and 73.4%), respectively (Fig. 2, Supplementary Fig. S1). No significant difference in local control was observed by tumor size (> 5 cm vs. ≤ 5 cm; 1-year rates, 96.9% vs. 96.6%; $p = 0.383$) or indication of radiotherapy (locally advanced primary tumor vs. oligometastasis; 1-year rates, 93.8% vs 100.0%; $p = 0.188$) (Supplementary Fig. S2). Over half of the patients (56.4%) developed subsequent distant progression. The 1-year OS was 88.6% and the 3-year OS was 61.4%, respectively. No statistically significant associations were found between local control or OS and patient age, sex, tumor size, treatment indication, or radiotherapy technique (Supplementary Tables S2, S3).

The crude incidences of grade 1–2 radiation pneumonitis and esophagitis were 46.7% and 8.1%, respectively. One patient developed grade 3 esophagitis that subsided 2 months after radiotherapy with supportive medications. In the radiotherapy plan for this index patient, approximately 5 cm length of the esophagus overlapped with the PTV. The maximal dose (Dmax) and mean dose (Dmean) of the esophagus were 42.4 Gy and 14.4 Gy, respectively.

Discussion and Conclusion

To the best of our knowledge, this study represents the first report of efficacy and safety data of radiotherapy for NSCLC using 40 Gy in 10 daily fractions. With a high overall response rate of 69.3%

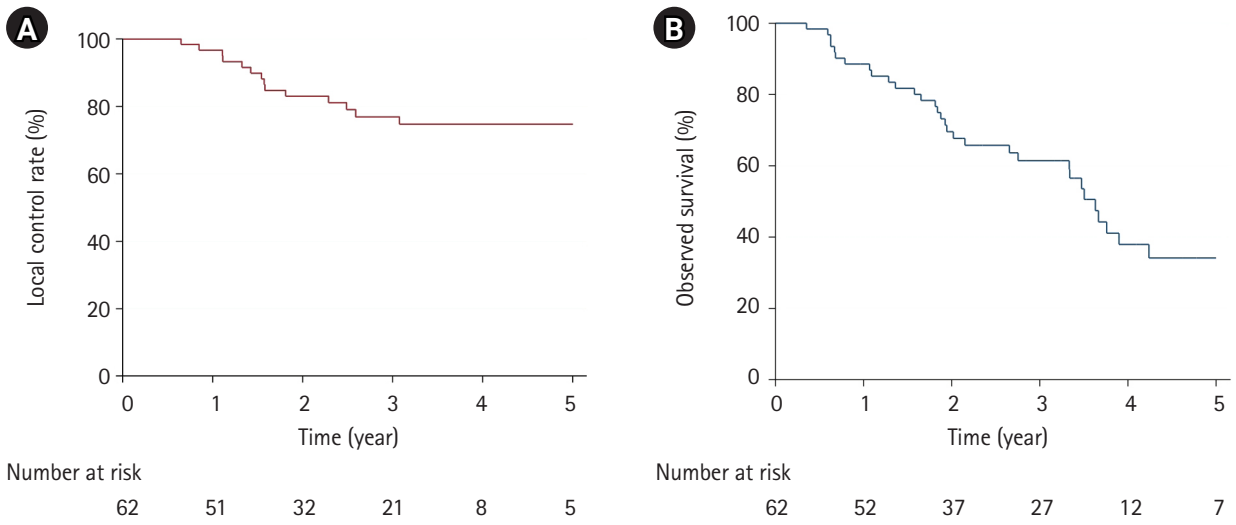

Fig. 2. (A) Local control rate and (B) overall survival of the cohort.

and an encouraging 1-year local control rate of 96.7%, this regimen represents a simple and effective radiotherapy scheme in attaining local tumor control.

A variety of dosing and fractionation schedules for palliative radiotherapy in NSCLC are used in clinical settings. Among the commonly employed regimens are 8–10 Gy in 1 fraction, 16–17 Gy in 2 fractions, 20 Gy in 5 fractions, 30 Gy in 10 fractions, and 36–39 Gy in 12–13 fractions [2]. The biologically effective dose (BED) 10, BED3, and equivalent doses in 2 Gy fractions of these regimens are detailed in Supplementary Table S4. Regarding palliative radiotherapy, the American Society for Radiation Oncology clinical practice guidelines recommend high-dose regimens (30 Gy in 10 fractions equivalent or greater) for NSCLC patients with good performance status [10]. This recommendation is consistent with other clinical guidelines and is primarily based on randomized data suggesting potential survival benefits with higher radiation doses compared to more modest treatment regimens [11]. In a meta-analysis of 13 randomized controlled trials, higher BED10 schedules of 35 Gy were associated with a greater likelihood of symptom improvement and longer survival compared to lower BEDs [5]. In this regard, the 40 Gy in 10 fractions regimen has a BED10 of 46.7 Gy, placing it at the higher end of the palliative thoracic radiotherapy dose spectrum.

Historically, studies evaluating conventional palliative radiotherapy regimens for NSCLC have primarily focused on symptom relief and OS as endpoints, the effect on local tumor control and toxicity outcomes was rarely reported [2,3]. The local control rate observed in our study closely aligns with data from other high-dose hypofractionated schedules for NSCLC, which were primarily investigated in the context of curative intent treatments. Using 3DCRT, Soliman et al. [12] reported long-term results of 118 patients with inoperable stage I–II NSCLC, treated with doses of 48–60 Gy delivered in 12–15 fractions, achieving a 5-year local control rate and OS of 70.1% and 40.0%, respectively. Comparable local control rates of 60%–70% have also been observed with more protracted schedules of 54–60 Gy in 18–20 fractions [13,14]. However, the incidence of grade ≥ 3 pneumonitis with the above schedules was reported to be up to 4–5%, necessitating corticosteroid treatment, and instances of fatal events have been documented. In this context, the clinical outcomes associated with the 40 Gy in 10 fractions regimen reported in the current study demonstrated a favorable balance between efficacy and safety, particularly in patients with suboptimal performance status. In addition to the satisfactory tumor response and control rates, the moderate hypofractionation over a brief 2-week treatment course is also convenient for patients and allows easy integration between cycles of systemic therapies.

The widespread adoption of immunotherapy and molecularly targeted agents has markedly improved the prognosis for patients with recurrent or metastatic NSCLC. Achieving sustained local tumor control with radiotherapy has become increasingly important for oligopersistent or oligoprogressive primary tumors following systemic treatments, as it serves to eradicate resistant tumor clones for potentially improved survival outcomes [15]. Although SBRT is widely recognized as the standard technique in these settings, its application to large or ultracentral tumors necessitates careful consideration of dosimetric feasibility and the potential for treatment-related toxicity. Several prospective studies have investigated SBRT approaches for centrally located NSCLC, reporting promising local control rates of 75%–90% [16-19]. However, the incidence of severe adverse events remain a concern, with grade ≥ 3 and grade 5 toxicities occurring in 7%–20% and 3%–15% of the treated pa-

tients, respectively. In the current cohort of patients treated by the 40 Gy in 10 fractions regimen, nearly one-third of the PTVs abutted the trachea and the proximal bronchial tree. Despite the ultracentral tumor location, no major hemorrhagic events were observed over a median follow-up duration of more than four years. Furthermore, the tumors included in this cohort were notably large, with a median PTV of 168.8 cm^3, corresponding to tumor diameters of approximately 6 to 7 cm. Traditionally, bulky tumors are considered more resistant to radiotherapy and are often underrepresented in SBRT trials. In one retrospective multicenter study of 92 patients with primary NSCLC measuring ≥5 cm treated with SBRT, local control rates at 1 and 2 years were 95.7% and 73.2%, respectively [20]. These outcomes closely align with the 1-year and 3-year local control rates of 96.7% and 76.9% observed in the present cohort, thereby supporting the efficacy of high-dose hypofractionated radiotherapy for achieving tumor control in large NSCLC.

The 40 Gy in 10 daily fractions regimen demonstrated a satisfactory safety profile in patients with NSCLC. The cohort primarily comprised an elderly population with fair performance status, as over half of the patients were aged 70 years or older. With all radiotherapy plans adhered to the pre-specified OAR constraints, the crude incidence of high-grade adverse event was low at 1.6%. Although pneumonitis was observed in 46.7% of patients, all cases were low-grade and primarily manifested as radiological changes in the absence of clinical symptoms. The only high-grade event was one case of grade 3 esophagitis, which occurred in a patient with a central lung tumor that abutted the esophagus, leading to a relatively high esophageal Dmax of 42.4 Gy. In the context of palliative thoracic radiotherapy for large tumors, where long segments of the esophagus may inevitably overlap with the PTV, prioritizing the esophagus during treatment plan optimization may further reduce the risk of symptomatic esophagitis while maintaining survival outcomes, especially when the prescribed radiotherapy dose is high [21]. This approach could be carefully considered in conjunction with the 40 Gy in 10 fractions regimen for patients in whom minimizing esophageal complication is a priority, allowing for a balance between effective tumor control and treatment-related toxicity.

This study has several limitations. First, while tumor response assessment was standardized using the RECIST criteria, accurate measurement of tumor size may be influenced by local tissue fibrosis caused by high-dose radiotherapy. Tumors surrounded by fibrotic scars can sometimes be mistakenly classified as RECIST-progressive diseases if only physical dimensions are considered. Functional imaging techniques, such as positron emission tomography, may provide additional information in evaluating responses in the post-radiation context, but it was not consistently available in our cohort [22]. As a result, our reported response and local control rates may be underestimated. Second, a switch in systemic therapy in any time-point after thoracic radiotherapy may confound response assessment and potentially lead to overestimation of tumor control. To address this, we conducted a separate sensitivity analysis by censoring this event, aiming to provide a more accurate estimation of our primary study endpoint. Third, due to the retrospective nature of this study, comprehensive data adverse events were not available. Certain toxicities, such as long-term chest wall and cardiac complications, were not routinely assessed hence may be underreported.

In conclusion, high-dose hypofractionated radiotherapy of 40 Gy in 10 daily fractions is an effective and safe regimen for local control of NSCLC. It represents a viable dose-fractionation with an aim for local response and control, especially in instances where tumor size or location renders SBRT unsuitable. Prospective evaluation of this regimen, along with careful comparisons to lower dose-fractionation schedules, is warranted to affirm its clinical value.

Statement of Ethics

This study protocol was reviewed and approved by the Central Institutional Review Board of the Hong Kong Hospital Authority (reference number: CIRB-2025-101-2). Waiver of informed consent was granted by the ethics committee due to the retrospective nature of the study.

Conflict of Interest

No potential conflict of interest relevant to this article was reported.

Funding

None.

Author Contributions

Conceptualization, JCHC (James C.H. Chow), JCHC (Jeffrey C.H. Chan), JYKC, GTCC, KMC, LTYL, KHW, KHA; Investigation and methodology, JCHC (James C.H. Chow), JCFL, CKCT, AHPT; Writing of the original draft, JCHC (James C.H. Chow); Reviewing and editing, all authors; Data curation, JCHC (James C.H. Chow), JCFL, CKCT, AHPT; Approval of final manuscript, all authors.

Data Availability Statement

The data that support the findings of this study are available from

the corresponding author upon reasonable request.

Supplementary Materials

Supplementary materials can be found via https://doi.org/10.3857/roj.2025.00346.

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